MIL-G-52732/12A
27 September 1985
SUPERSEDING
MIL-G-52732/12
23 June 1972

MILITARY SPECIFICATION SHEET

GENERATOR SET, GASOLINE ENGINE DRIVEN, 10 KW, 400 HERTZ

TYPE I (TACTICAL), CLASS 2 (UTILITY)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the generator set described herein shall consist of this document and the issue in effect of specification MIL-G-52732.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for mode II, size 10 sets, model MEP-023A.

1.2 Classification.

1.2.1 Set rating. The mode II set shall be rated 0.8 power factor (PF) and shall be connectable for the following voltage ratings:

a. 1 phase, 2 wire, 120 volts.
b. 1 phase, 3 wire, 120/240 volts.
c. 3 phase, 3 wire, 120 volts.
d. 3 phase, 4 wire, 120 volts line-to-neutral and 208 volts line-to-line (120/208 volts).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6115

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SPECIFICATIONS

MILITARY

MIL-H-52112	- Heating Torches, Equipment Winterization, Instant Lighting.
MIL-G-52732	- Generator Sets, Gasoline Engine Driven, 0.5 KW thru 10 KW, 60 Hertz, 400 Hertz and 28 Volt Direct Current, Type I (Tactical), Class 2 (Utility) General Specification for.

STANDARDS

MILITARY

MS39302	- Engine, Gasoline; Air Cooled, Industrial-Type, 4-cycle, 20 Net Continuous HP.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

ME

TA13216E1900	Generator Set, Gas-Eng-Driven, 10.0 KW, 400 Hz, AC, Air-Cooled, Portable, Tubular-Frame, Skid-Mounted.
TA13218E8060	Winterization Kit, 10 KW, 400 Hz, Generator Set.

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specifications sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Requirements shall be in accordance with MIL-G-52732, and as specified herein.

3.2 Description. The set shall be in accordance with Top Assembly TA13216E1900 and as specified herein.

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3.3 Starting and operating. Servicing, adjustment and replacement shall be as specified in table I.

3.3.1 Without winterization. The remote start-stop provisions are applicable for the mode II, size 10 set.

3.3.2 With winterization. At ambient temperatures from -25° F (-31.7° C) to -65° F (-53.9° C), the set shall start within 15 minutes with the application of external heat from a torch conforming to MIL-H-52112, type I. The set shall meet these requirements utilizing a winterization kit in accordance with TA13218E8060.

3.4 Maximum power. The maximum power (corrected) in kilowatts of the set shall be not less than 18.5 kW when operated at rated speed, voltage and at unity power factor.

3.5 Engine.

3.5.1 Engine design. The Government-furnished engines will conform to MS39302.

3.5.2 Engine temperature. Engine temperatures shall not exceed the following:

Location	Maximum Temperatures	
	First 10 hours of engine operation	After 10 hours of engine operation
Sump oil.	300° F (148.9° C)	265° F (129.2° C)
Cylinder head.	500° F (260° C)	475° F (246.1° C)
Exhaust.	1200° F (648.9° C)	1200° F (648.9° C)

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TABLE I. Schedule of servicing and adjusting.

All items may be serviced before test.

Item No.	Hours between service periods	Item
1	8	Any minor maintenance such as retightening and retorquing nuts, bolts and screws that can be performed without disturbing or removing adjacent components, accessories or shrouding (except top shroud, pump, intake manifold and exhaust system).
2	Not allowed	Any major maintenance that requires removal of other components, accessories or shrouding such as oil pans, cylinder heads and connecting rods.
3	As required	Cleaning and regapping or replacing spark plugs.
4	As required	Servicing fuel filter.
5	100	Checking compression pressures.
6	8	Checking and adding lubricating oil, as necessary.
7	8	Adjusting carburetor.
8	As required	Cleaning carburetor.
9	8	Adjusting governor (normal speed adjustment).
10	100	Changing lubricating oil.
11	100	Changing lubricating oil filter element.
12	100	Servicing air cleaner.
13	300	Adjusting and dressing breaker points.
14	300	Adjusting ignition timing.
15	1500	Cleaning and inspection of combustion chamber, manifolds and valves.
16	500	Replacing breaker points.
17	As required	Replacing condenser.
18	*	Gasket replacement.

* Not allowed except where component is removed for servicing.

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3.5.3 Governing system.

3.5.3.1 Short term, steady-state stability (30 seconds), mode II and III sets. The short term steady-state stability bandwidth percentage shall not exceed 1 percent.

3.5.3.2 Long term, steady-state stability (4 hours), mode II and III sets. The long term steady-state stability bandwidth percentage shall not exceed 2 percent.

3.6 Generator.

3.6.1 Voltage unbalance, mode II and III sets. The variation between line-to-line voltages shall not exceed 5 percent.

3.6.2 Phase balance (voltage), mode II and III sets. The maximum difference in line-to-line and line-to-neutral voltages shall not exceed 1 percent.

3.6.3 Voltage waveform, mode II and III sets. Single frequency harmonics shall not exceed 2.0 percent. For the 120/240 volt, 1-phase, 3-wire connection, this requirement applies when measured at the convenience receptacle. The deviation factor of the line-to-neutral and line-to-line voltages for the 3-phase generator connections at rated load, no-load and unity power factor rated kW shall not exceed 5.0 percent. Under the same conditions the deviation factor at the single phase connections shall not exceed 6.0 percent.

3.6.4 Output voltage ripple, mode IV sets. Not applicable for the mode II, size 10 set.

3.6.5 Saturation curves. With the generator temperature stabilized at normal ambient conditions the exciter field current for rated voltage shall not exceed 0.42 ampere at no-load and 0.55 ampere at rated load.

3.6.6 Voltage adjustment range. The minimum acceptable voltage adjustment range shall be 5 percent above and below rated voltage.

3.6.7 Voltage regulation. The voltage regulation percentage shall not exceed 3 percent.

3.6.8 Voltage steady-state performance. The voltage steady-state stability bandwidth percentage shall not exceed 1 percent.

3.6.9 Voltage transient performance. The voltage dip or rise percentage shall not exceed 20 percent.

3.6.10 Generator efficiency. The generator efficiency shall not be less than 84 percent at 1.0 PF and 81 percent at 0.8 PF.

3.7 Circuit interrupter. The circuit interrupter shall open within 1.0 seconds with 200 percent load applied.

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3.8 Instruments. The inaccuracy or error of the metering system shall not exceed the following:

Instrument	Allowable Value
Ammeter (percent load)	10 percent inaccuracy
Voltmeter	3 percent inaccuracy
Frequency Meter	5 Hz error

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance provisions. The quality assurance provisions shall be in accordance with MIL-G-52732 and as specified herein.

4.2 Engine preventative maintenance services. The engine preventative maintenance services shall be performed as specified in table I.

4.3 Drop test (free fall). The height of the drop shall be 12 inches for the mode II, size 10 set.

4.4 Vibration test. The hand rail or shock mount resonances shall not be utilized for resonant dwell conditions.

5. PACKAGING

5.1 Preparation for delivery. Preparation for delivery shall be in accordance with MIL-G-52732.

6. NOTES

6.1 Notes. The notes specified in MIL-G-52732 are applicable to this specification and as specified herein.

6.2 Stable operation (voltage). Stable operation is operation within a bandwidth of 1 percent of the rated voltage.

6.3 International standardization. Not applicable for the mode II, size 10 set.

Custodians:
Army - ME
Air Force - 99
Navy - YD

Preparing activity:
Army - ME

Project 6115-0240

Review activities:
Army - ER, GL
Air Force - 80

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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

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